

OE-EC804B — Microwave Integrated Circuits

Semester VIII • 3L:0T:0P • 3 Credits

Module 1 Exam Notes: Introduction to MIC & MMIC

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Contents

1	Introduction to MIC and MMIC	2
1.1	Classification: Hybrid MIC (HMIC) vs. Monolithic MIC (MMIC)	2
1.2	Advantages of MICs over Discrete & Waveguide Circuits	2
2	MMIC Fabrication Techniques and Materials	3
2.1	Substrate Materials for MMICs	3
2.2	Thin Film Technology (Film Thickness < 5 μm)	3
2.3	Thick Film Technology (Film Thickness 10 – 50 μm)	4
3	Encapsulation and Mounting of Active Devices	4
3.1	Active Device Mounting Techniques	4
3.2	Encapsulation and Package Requirements	5

1 Introduction to MIC and MMIC

Microwave Integrated Circuits (MICs) operate in the high-frequency spectrum ranging from 300 MHz to 300 GHz (wavelengths $\lambda = 1\text{ m}$ to 1 mm). Conventional low-frequency lumped elements (discrete resistors, capacitors, and inductors) fail at microwave frequencies due to parasitic lead inductances and stray capacitances. MIC technology replaces bulky 3D waveguide structures with planar transmission lines printed directly on insulating or semiconductor substrates.

Microwave Integrated Circuit (MIC)

A **Microwave Integrated Circuit (MIC)** is a high-frequency planar assembly where passive circuit elements (printed microstrip lines, inductors, capacitors, couplers) and active semiconductor devices (diodes, MESFETs, HEMTs, HBTs) are integrated on a common planar substrate.

1.1 Classification: Hybrid MIC (HMIC) vs. Monolithic MIC (MMIC)

MICs are categorized into two major technological generations:

- Hybrid MIC (HMIC):** Passive matching networks and transmission lines are fabricated on a passive dielectric substrate (such as Alumina Al_2O_3 , Quartz, or Sapphire) using thin or thick film deposition. Active semiconductor devices (packaged or unpackaged GaAs/Si FET dies, Schottky/PIN diodes) are individually mounted onto the substrate and connected using gold wire bonds or ribbon bonds.
- Monolithic MIC (MMIC):** All active semiconductor devices (MESFETs, HEMTs, HBTs), passive matching components (MIM capacitors, thin film resistors, spiral inductors), and planar interconnect transmission lines are grown, etched, and metallized **simultaneously on a single semiconductor wafer** (such as Gallium Arsenide GaAs, Indium Phosphide InP, or Gallium Nitride GaN).

Feature	Hybrid MIC (HMIC)	Monolithic MIC (MMIC)
Substrate Material	Dielectric (Alumina Al_2O_3 , Quartz, Sapphire, PTFE)	Semi-insulating Semiconductor (GaAs, InP, GaN, SiGe)
Active Device Integration	Discrete semiconductor dies mounted & wire-bonded individually	Fabricated monolithically directly inside/on the semiconductor wafer
Parasitic Reactance	High due to wire bond inductances ($L_s \approx 0.5 - 2\text{ nH}$)	Negligible; planar interconnects eliminate bond wire inductances
Frequency Range	Typically $< 20\text{ GHz}$ due to bond wire parasitics	Millimeter-wave up to 300 GHz and THz frequencies
Production Volume	Ideal for prototype development and low-volume production	Highly economical for mass batch production ($10^4 - 10^7$ units)
Unit Cost	Low initial mask cost; high unit assembly labor cost	High initial mask/NRE cost; extremely low per-chip unit cost
Post-Fabrication Tuning	Trimming of capacitors/resistors possible post-fabrication	No post-fabrication tuning possible; design must be 100% accurate
Thermal Dissipation	Excellent thermal transfer through thick ceramic substrates	Moderate to poor; requires thermal via holes through wafer

1.2 Advantages of MICs over Discrete & Waveguide Circuits

- Miniaturization & Weight Reduction:** Planar integration reduces volume and weight by over 90% compared to metallic waveguides.
- High Reliability & Reproducibility:** Batch photolithography eliminates mechanical assembly variations and hand-soldered joints.
- Broadband Performance:** Short planar interconnects minimize parasitic leads, extending operating bandwidths to multi-octave ranges.

- **Multifunctional Systems-on-Chip (SoC):** Enables complete transmit/receive (T/R) radar modules on a single die.

2 MMIC Fabrication Techniques and Materials

2.1 Substrate Materials for MMICs

The choice of substrate dictates electrical attenuation, substrate power loss, high-frequency cutoff, and power dissipation capabilities.

Substrate Requirements for MMICs

An ideal MMIC substrate must exhibit:

1. High electrical resistivity ($\rho > 10^7 \Omega \cdot \text{cm}$) to minimize RF substrate conduction losses.
2. Low dielectric loss tangent ($\tan \delta < 0.001$) for minimal wave attenuation.
3. High thermal conductivity (K) to dissipate heat from high-power active devices.
4. High electron mobility (μ_e) to enable high-speed transistor switching.

Substrate	Dielectric Constant (ϵ_r)	Loss Tangent ($\tan \delta$)	Thermal Cond. ($W/cm \cdot K$)	Key Properties & Applications
Gallium Arsenide (GaAs)	12.9	0.0006	0.46	Standard workhorse for MMICs up to 100 GHz; high mobility ($\mu_e = 8500 \text{ cm}^2/\text{V} \cdot \text{s}$).
Indium Phosphide (InP)	12.4	0.0010	0.68	Sub-millimeter wave and THz MMICs ($> 100 \text{ GHz}$); ultra-high peak electron velocity.
Gallium Nitride (GaN)	8.9	0.0005	1.30	High-power microwave amplifiers; high breakdown field (3 MV/cm).
Silicon (Si / High-Res)	11.7	0.0050	1.45	RF-CMOS and SiGe HBT integration; cheap mass production up to 60 GHz.
Alumina (Al_2O_3)	9.8	0.0001	0.30	Standard substrate for Hybrid MICs; excellent dielectric stability.

2.2 Thin Film Technology (Film Thickness $< 5 \mu\text{m}$)

Thin film technology is used in all high-frequency MMICs and precision Hybrid MICs.

- **Deposition Methods:**
 - *Vacuum Evaporation:* Thermal or electron-beam heating vaporizes metal atoms in high vacuum (10^{-6} Torr).
 - *RF Sputtering:* Argon ions bombard a target material, knocking off atoms that deposit uniformly onto the substrate.
 - *Electroplating:* Electro-chemical growth used to thicken conductor traces to several skin depths (2 – 5 μm of Gold Au).

- **Passive Component Fabrication:**

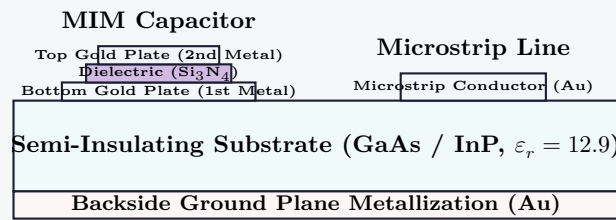
- *Thin Film Resistors:* Tantalum Nitride (TaN) or Nickel Chromium (NiCr) providing sheet resistivities of $10 - 100 \Omega/\text{sq}$.
- *Metal-Insulator-Metal (MIM) Capacitors:* Top and bottom Gold plates separated by a thin dielectric film of Silicon Nitride (Si_3N_4 , $\epsilon_r = 7.5$) or Silicon Dioxide (SiO_2 , $\epsilon_r = 3.9$).

2.3 Thick Film Technology (Film Thickness $10 - 50 \mu\text{m}$)

Used primarily for low-frequency, high-power Hybrid MICs below 10 GHz.

- Deposited via **Screen Printing** of conductive or resistive pastes through fine stainless-steel mesh screens onto ceramic substrates.
- Fired in high-temperature belt furnaces at 850°C to sinter metal particles.
- Minimum resolution is limited to $> 75 \mu\text{m}$ line widths.

Cross-Sectional Structure of Thin-Film MIM Capacitor and Microstrip Line on GaAs



MIM Capacitor Calculation

Problem: Calculate the required area of a Silicon Nitride (Si_3N_4 , $\epsilon_r = 7.5$) MIM capacitor to achieve a capacitance of $C = 10 \text{ pF}$ if the dielectric thickness is $d = 2000 \text{ \AA}$ (200 nm).

Solution: Using the parallel-plate capacitance formula:

$$C = \frac{\epsilon_0 \epsilon_r A}{d} \Rightarrow A = \frac{C \cdot d}{\epsilon_0 \epsilon_r} \quad (1)$$

Given: $C = 10 \times 10^{-12} \text{ F}$, $d = 200 \times 10^{-9} \text{ m}$, $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$, $\epsilon_r = 7.5$.

$$\begin{aligned} A &= \frac{(10 \times 10^{-12}) \times (200 \times 10^{-9})}{(8.854 \times 10^{-12}) \times 7.5} \\ &= \frac{2 \times 10^{-18}}{6.6405 \times 10^{-11}} = 3.0118 \times 10^{-8} \text{ m}^2 = 0.0301 \text{ mm}^2 \end{aligned} \quad (2)$$

For a square capacitor plate of length L :

$$L = \sqrt{A} = \sqrt{0.0301 \text{ mm}^2} \approx 0.1735 \text{ mm} = 173.5 \mu\text{m} \quad (3)$$

3 Encapsulation and Mounting of Active Devices

3.1 Active Device Mounting Techniques

In Hybrid MICs, discrete active semiconductor chips (diodes, MESFETs, HBTs) must be attached to the substrate and interconnected to microstrip lines. Three primary mounting techniques are utilized:

1. Wire Bonding (Thermosonic / Ultrasonic):

- Fine Gold (Au) wires ($17.5 - 25 \mu\text{m}$ diameter) connect die pads to substrate traces.
- *Drawback:* Introduces parasitic series inductance $L_s \approx 1 \text{ nH/mm}$. At 20 GHz, a 1 mm wire bond introduces an inductive reactance $X_L = 2\pi f L_s = 2\pi(20 \times 10^9)(10^{-9}) = 125.66 \Omega$, destroying impedance matching!

2. Flip-Chip Bonding:

- The semiconductor die is inverted face-down and attached directly to substrate pads via solder bumps or gold pillars ($20 - 50 \mu\text{m}$ height).
- Reduces parasitic series inductance by over 95% ($L_s < 0.05 \text{ nH}$), enabling performance up to 100 GHz.

3. Tape Automated Bonding (TAB):

- Patterned copper foil leads on a flexible polyimide film are gang-bonded simultaneously to die pads.

3.2 Encapsulation and Package Requirements

Packaging protects delicate GaAs/InP devices from environmental humidity, oxidation, and mechanical stress while providing a low-reflection 50Ω RF path.

Hermetic Packaging

Hermetic Packaging encloses MIC/MMIC circuits inside glass-sealed ceramic or laser-welded metal housings filled with dry nitrogen, preventing moisture ingress and long-term performance degradation.

Quick Revision — Module 1

Introduction to MIC & MMIC | OE-EC804B

Key Concepts and Metrics at a Glance

Hybrid MIC (HMIC)	Passives on dielectric substrate (Al_2O_3); discrete active dies wire-bonded.
Monolithic MIC (MMIC)	Passives, active devices, and lines fabricated monolithically on single GaAs/InP wafer.
Substrate Materials	GaAs ($\epsilon_r = 12.9, \mu_e = 8500$), InP ($\epsilon_r = 12.4$), GaN ($\epsilon_r = 8.9, 3 \text{ MV/cm}$).
MIM Capacitance	$C = \frac{\epsilon_0 \epsilon_r A}{d}$ (Si_3N_4 dielectric, $\epsilon_r = 7.5$).
Wire Bond Parasitic	$L_s \approx 1 \text{ nH/mm} \Rightarrow X_L = 2\pi f L_s$; mitigated by Flip-Chip bonding ($< 0.05 \text{ nH}$).

One-Line Definitions

- **MMIC:** An integrated circuit containing all active and passive microwave components on a single semiconductor chip.
- **MIM Capacitor:** Metal-Insulator-Metal thin film capacitor providing high capacitance density in MMICs.
- **Loss Tangent ($\tan \delta$):** Ratio of imaginary to real permittivity (ϵ''/ϵ') measuring dielectric RF dissipation.
- **Flip-Chip Bonding:** Surface-mount method bonding an inverted chip face-down via solder bumps, eliminating wire bond inductance.

Frequently Asked / Viva Questions

Q1. Compare Hybrid MICs and Monolithic MICs.

HMICs use separate dielectric substrates with individually mounted active chips. MMICs fabricate all active and passive components simultaneously on a single semiconductor substrate (GaAs/InP).

Q2. Why is GaAs preferred over Silicon for MMICs?

GaAs is a direct bandgap semiconductor with semi-insulating resistivity ($\rho > 10^7 \Omega \cdot \text{cm}$) and higher electron mobility ($\mu_e \approx 8500 \text{ cm}^2/\text{V} \cdot \text{s}$ vs 1400 for Si), providing low RF substrate losses.

Q3. What are the advantages of MMICs over discrete microwave circuits?

Small size, low weight, high reliability, high reproducibility, elimination of wire bond parasitics, and low mass-production cost.

Q4. Explain Thin Film technology in MIC fabrication.

Thin film technology deposits metal and dielectric films ($< 5 \mu\text{m}$) via sputtering or evaporation, using photolithography to achieve high-precision line widths down to sub-micron scales.

Q5. What materials are used for thin film resistors and capacitors?

Resistors use Tantalum Nitride (TaN) or Nickel Chromium (NiCr). Capacitors use Metal-Insulator-Metal (MIM) with Si_3N_4 or SiO_2 dielectrics.

Q6. What is the main electrical drawback of wire bonding at microwave frequencies?

Wire bonds introduce parasitic series inductance ($L_s \approx 1 \text{ nH/mm}$), causing significant impedance mismatches above 12 GHz.

Q7. How does Flip-Chip bonding overcome wire bond parasitics?

Frequently Asked / Viva Questions (contd.)

Flip-Chip bonding mounts the chip face-down directly onto substrate pads using short solder bumps, reducing parasitic inductance by over 90%.

Q8. Why is high thermal conductivity critical for MMIC substrate materials?

High-power active devices generate significant heat in a small area; high thermal conductivity prevents thermal breakdown and maintains transistor gain.

Q9. Define loss tangent ($\tan \delta$) and its significance in MICs.

Loss tangent $\tan \delta = \epsilon''/\epsilon'$ quantifies dielectric power loss; lower $\tan \delta$ minimizes attenuation in planar transmission lines.

Q10. What is Hermetic Sealing and why is it necessary for MMICs?

Hermetic sealing provides an airtight ceramic/metal package protecting unpassivated GaAs active devices from moisture and atmospheric degradation.